Diffusion-Guided Autoregressive Models for Preserving Bragg Peaks in XRD Modeling

Yarong Li
ICME
Stanford University
yarongli@stanford.edu

Juan Muneton Gallego
ICME
Stanford University
jmuneton@stanford.edu

Cong Wang
SLAC
Stanford University
cwang31@slac.stanford.edu

1 Introduction

In structural biology and chemistry, uncovering the three-dimensional structure of molecules is essential for advancing our understanding of biological mechanisms, drug design, and materials science. However, molecules are exceedingly small—on the order of 0.1 to 0.3 nanometers—and cannot be visualized using conventional light microscopy. Instead, scientists rely on *X-ray crystallography*, a powerful technique that probes atomic-scale structures using the diffraction patterns generated when X-rays interact with crystalline samples. Landmark discoveries, such as the double-helix structure of DNA, were made possible through this method.

At the SLAC National Accelerator Laboratory, the *Linac Coherent Light Source* (LCLS) is a state-of-the-art facility that produces ultrafast X-ray laser pulses capable of capturing high-resolution diffraction images. With its upcoming upgrade, LCLS will increase its pulse rate from 120 to nearly one million pulses per second. This dramatic boost in data acquisition will vastly expand our ability to explore molecular structures—but it also introduces a severe data bottleneck. Each X-ray pulse produces a high-resolution diffraction image, often reaching sizes of 1920×1920 pixels with high bit-depth, resulting in immense computational and storage demands.

This surge in data volume presents a significant challenge for downstream machine learning tasks. In particular, high-resolution images are inherently high-dimensional, and the computational cost of training models scales with input size. As LCLS advances into the exascale regime, there is a critical need for efficient models that can learn the underlying structure of these images while preserving key scientific features.

Our team at SLAC addresses this challenge by developing machine learning-based image compression methods that enable compact representations of diffraction images without sacrificing scientific fidelity. The central objective is to preserve two crucial structural features:

- **Bragg Peaks** – Sparse, high-intensity points that encode fine-grained crystalline structure.
- **Coarse Ring Structures** – Broader radial patterns indicative of large-scale molecular organization.

By focusing on models that retain both local and global image structures, we aim to ensure that compressed representations remain useful for downstream scientific analysis. In this work, we investigate the effectiveness of *Variational Autoencoders*, *Vector-Quantized Autoencoders*, and *autoregressive diffusion models* in learning the underlying data distribution and reconstructing images that preserve key scientific features. These models offer complementary approaches to compression and generation, allowing us to evaluate how well each can capture and retain critical characteristics such as Bragg peaks and coarse ring patterns.

2 Dataset

Our dataset is composed of real experimental X-ray diffraction data collected at the *Linac Coherent Light Source* (LCLS), encompassing two large-scale experiments conducted using distinct detector configurations. Each experiment is subdivided into multiple runs, where each run contains a fixed number of diffraction images. Specifically:

- The first experiment consists of 300 runs, also with 40 images per run (Labeled as Experiment 422).
- The second experiment includes 179 runs, each containing 40 images (Labeled as experiment 522).

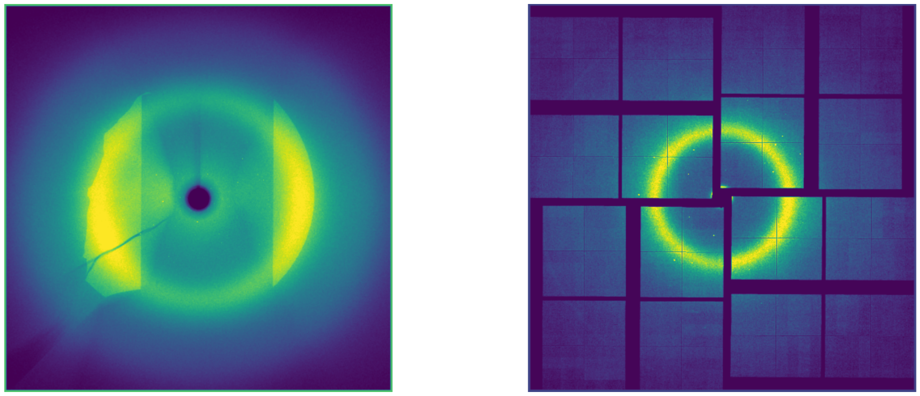


Figure 1: Side-by-side comparison of two sample images from two different detectors. On the left, we have the detector from experiment 422 and on the right we have detector from experiment 522

Each image is a single-channel (grayscale) diffraction pattern with a resolution of 1920×1920 pixels and a bit depth ranging from 14 to 16 bits, allowing for the representation of high-dynamic-range intensities.

2.1 Data Pre-processing

To prepare the diffraction images for training, we apply the following preprocessing steps to ensure stability during optimization and to focus learning on scientifically meaningful features:

1. **Standardization:** Pixel intensities are normalized (zero mean, unit variance) to stabilize gradient updates and improve convergence during training.
2. **Edge Masking:** The outer 10-pixel boundary on all four edges of each image is masked out to eliminate extreme intensity values caused by detector artifacts. This encourages the model to focus on the central region where meaningful diffraction patterns and Bragg peaks typically appear.
3. **Downsampling via Average Pooling:** To reduce computational overhead, images are spatially downsampled using average pooling. This preserves coarse structural information—such as ring patterns and peak densities—while significantly lowering input dimensionality. Note that we conducted experiments using down sampling and without down sampling to better understand the trade-offs of image sampling against computational requirements.

3 Models

3.1 Variational Autoencoder (VAE)

Variational autoencoders (VAEs), originally introduced by [1], are a class of probabilistic generative models that learn compact, lower-dimensional representations of high-dimensional data. Unlike traditional autoencoders, which deterministically map inputs to compressed representations, VAEs model the latent space as a probability distribution, allowing for both dimensionality reduction and generative sampling.

At a high level, a VAE consists of two neural networks: an *encoder* $h_\phi(x)$ that maps the input x to a latent representation z, and a *decoder* $f_\theta(z)$ that reconstructs the original input, producing $\hat{x}$, as shown in Figure 2. A key distinction of the VAE is that the encoder outputs parameters of a probability distribution (typically the mean and log-variance of a Gaussian), from which the latent variable z is sampled using the reparameterization trick to ensure differentiability during training.

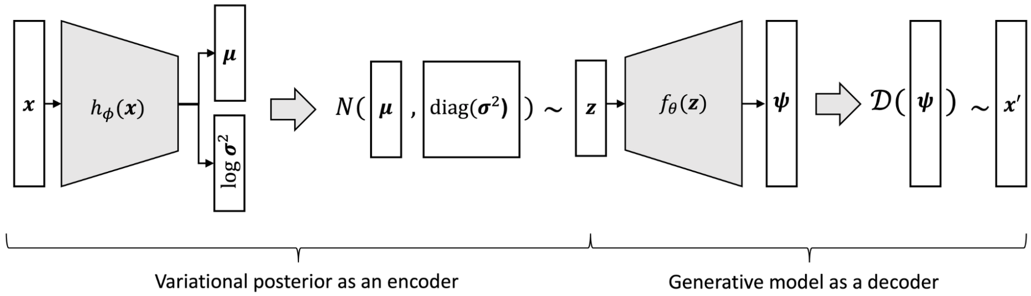


Figure 2: Architecture of a Variational Autoencoder (VAE). The encoder network $h_\phi(x)$ outputs mean and log-variance vectors used to sample latent variable z, which is decoded by $f_\theta(z)$ and further processed to reconstruct x'.

Formally, VAEs are trained to maximize a variational lower bound on the marginal log-likelihood of the data, which balances two terms: a reconstruction loss and a regularization term encouraging the learned latent distribution to be close to a prior (usually a standard normal distribution). This objective can be written as:

$$\mathcal{L}(\phi, \theta; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x) \| p(z))$$

Early in training, the KL term can dominate, leading to *posterior collapse*, where the latent space becomes uninformative. KL annealing mitigates this by gradually increasing a weighting factor $\beta(t)$ on the KL term, allowing the model to first focus on reconstruction before enforcing regularization. This promotes stable training and more meaningful latent representations. The annealed objective is:

$$\mathcal{L}_{\text{annealed}}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta(t) \cdot D_{\mathrm{KL}}(q_\phi(z|x) \parallel p(z))$$

VAEs can be understood from two complementary perspectives:

1. **Probabilistic Generative Model:** VAEs model the data generation process by introducing latent variables z for each input x, and jointly modeling $p(x, z)$. This enables generation of new data by sampling from the prior $p(z)$ and decoding it using $f_\theta(z)$.
2. **Probabilistic Autoencoder:** As autoencoders, VAEs map each input to a distribution over the latent space rather than a fixed point. This probabilistic formulation allows for more expressive representations and improved generalization in downstream tasks.

In this work, we explore the use of VAEs for modeling high-resolution X-ray diffraction data, leveraging their ability to capture underlying structure while enabling compact, informative latent representations.

3.2 Vector-Quantized Variational Autoencoder (VQ-VAE)

The *Vector-Quantized Variational Autoencoder (VQ-VAE)* [2] is a variant of the variational autoencoder that learns discrete latent representations by incorporating a vector quantization step into the bottleneck of the autoencoder. Unlike standard VAEs, which model the latent space as continuous and Gaussian-distributed, VQ-VAE constrains the latent space to a finite set of learnable embedding vectors—also known as a *codebook*.

VQ-VAE differs from standard VAEs in two important ways:

- The encoder does not output continuous latent variables, but instead maps each input to the nearest embedding vector from a fixed codebook, effectively discretizing the latent space.

- The prior over the discrete latent codes is learned (e.g., via an autoregressive model), rather than being fixed as a standard Gaussian.

This discretization process addresses the problem of *posterior collapse*, a common issue in VAEs where the decoder becomes so powerful that it ignores the latent variables entirely. By enforcing a discrete bottleneck, the VQ-VAE ensures that meaningful latent representations must be learned and used by the decoder.

During training, the encoder maps an input x to a continuous latent vector $z_e(x) \in \mathbb{R}^d$. This vector is then replaced with the nearest codebook entry e_k from a learned set of embeddings $\{e_i\}_{i=1}^K$, where K is the codebook size. Formally, the quantized latent vector is:

$$z_q(x) = \mathrm{sg}\left(e_k\right), \quad \text{where} \quad k = \arg\min_i \|z_e(x) - e_i\|_2$$

and `sg` denotes the stop-gradient operator, which prevents gradients from flowing into the codebook selection step.

The decoder then reconstructs the input as $\hat{x} = f_\theta(z_q(x))$. The model is trained by optimizing the following loss:

$$\mathcal{L} = \underbrace{\|x - \hat{x}\|^2}_{\text{Reconstruction Loss}} + \underbrace{\|\mathrm{sg}[z_e(x)] - e_k\|^2}_{\text{Codebook Loss}} + \underbrace{\beta\|z_e(x) - \mathrm{sg}[e_k]\|^2}_{\text{Commitment Loss}}$$

The reconstruction loss ensures that the output matches the input. The codebook loss updates the codebook entries to move toward the encoder outputs, and the commitment loss encourages encoder outputs to stay close to their assigned codebook vectors.

3.3 Masked Autoregression Model (MAR)

The Masked Autoregressive Model (MAR) [3] is a Transformer-based encoder–decoder architecture designed to model images or sequences via masked generation. Both the encoder and decoder share the same architecture, typically consisting of 32 Transformer blocks with a hidden width of 1024. The encoder receives only a subset of visible tokens along with 64 prepended `[cls]` tokens, which stabilize training and improve the capacity of the encoder. These tokens are processed using self-attention, yielding a latent representation. The decoder, which mirrors the encoder in depth and dimensionality, is fed the masked token positions (represented by `[m]` tokens) and leverages cross-attention to condition on the encoder's output. To maintain positional consistency across inputs, the model uses absolute positional embeddings for both visible and masked tokens.

During training, MAR adopts a masked autoregressive learning scheme in which a random proportion of tokens are masked using a uniformly sampled masking ratio from the range $[0.7, 1.0]$. Unlike classical autoregressive models that predict tokens in a fixed order, MAR performs *next set-of-tokens prediction* by dividing the token sequence into random subsets and autoregressively predicting one subset at a time. This is done via bidirectional attention, allowing the model to reason over known tokens when predicting each new group of unknowns. During decoding, masked token positions are filled using shared attention layers and positional information, allowing MAR to flexibly reconstruct high-dimensional outputs even under sparse supervision.

The loss function of MAR follows the denoising objective from standard diffusion models but is conditioned on the autoregressive context. Specifically, at each step, the MAR model provides a conditioning vector $z = f(\cdot)$ for the diffusion process. The diffusion model then predicts the noise ε added to the ground truth token x at time step t, conditioned on z. The objective is to minimize the expected squared error between the true noise and the predicted noise:

$$\mathcal{L}(z, x) = \mathbb{E}_{\varepsilon, t}\left[\|\varepsilon - \varepsilon_\theta(x_t \mid t, z)\|^2\right],$$

where $x_t = \sqrt{\bar{\alpha}_t}x + \sqrt{1 - \bar{\alpha}_t}\varepsilon$ is the noised input, and ε_θ is a noise estimator parameterized by θ. This formulation tightly couples the autoregressive model and the diffusion model: the former provides a latent context z used to condition the denoising process, while the latter learns to model the conditional distribution $p(x \mid z)$. Conceptually, this allows MAR to guide the diffusion model in generating plausible token values from partially observed inputs, enabling accurate and context-aware reconstruction.

At inference time, MAR generates outputs progressively by reducing the masking ratio from 1.0 to 0 using a cosine decay schedule. In each step, a new subset of tokens is selected at random and predicted using the model's current state. This allows MAR to predict multiple tokens simultaneously rather than committing to a single autoregressive ordering. To introduce diversity and robustness in predictions, a temperature-based sampling strategy (τ) is applied to the model's output probabilities.

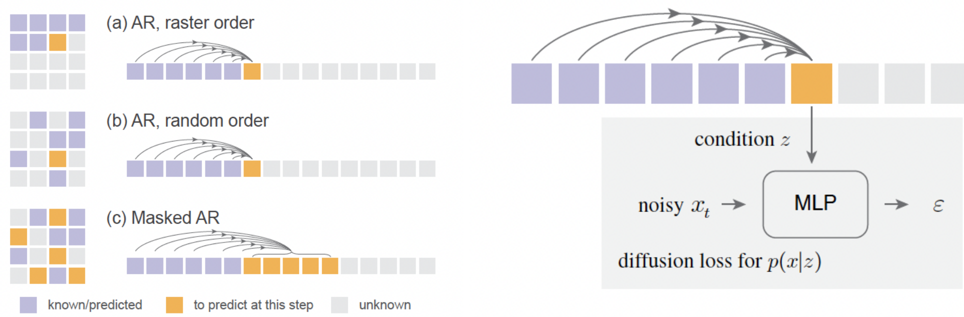


Figure 3: Left: Illustration of the autoregressive strategy used by MAR (c), where image patches are treated as tokens and the model predicts the next set of patches conditioned on the known ones. Right: The full MAR architecture, which employs a transformer-based model to learn latent embeddings z. These embeddings, combined with a forward diffusion process, are used to predict the noise parameter ϵ required for the reverse diffusion process that generates new tokens.

We followed the training approach presented in [4], where a pre-trained VAE is first used to encode the original dataset into a latent space. The resulting latent vectors are then passed through a diffusion process, and the final latent obtained from the reverse diffusion is decoded back into image space. As shown in Figure 4, this design is motivated by the need to balance generative quality with computational efficiency. Applying diffusion models directly in high-dimensional pixel space is computationally intensive and often requires a large number of denoising steps to achieve high-fidelity outputs. By leveraging a VAE encoder to compress images into a lower-dimensional, semantically meaningful latent space, the diffusion process can operate more efficiently and with a greater focus on high-level structure rather than low-level pixel noise. Finally, a decoder reconstructs the image from the refined latent variables, preserving detail while benefiting from the more structured and tractable generative modeling in latent space.

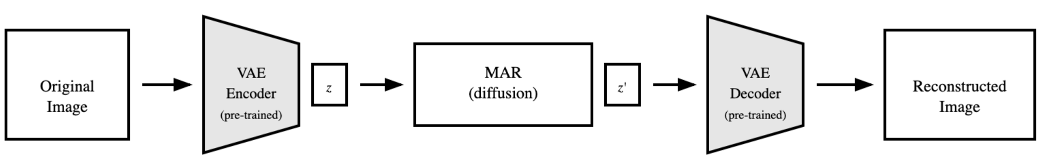


Figure 4: Overview of the MAR training pipeline. The original image is first encoded into a latent representation z using a pre-trained VAE encoder. The latent is then passed through the MAR model, which applies a diffusion process to produce a refined latent z'. Finally, the pre-trained VAE decoder reconstructs the image from z'. This architecture enables efficient training of the autoregressive model in latent space while preserving high-quality reconstructions.

4 Results

4.1 VAE

All models were trained for 10 epochs with a batch size of 2 using a learning rate of 1×10^{-5} and a weight decay of 0.001. We explored several reconstruction loss functions to guide training, including the standard ℓ_1 loss, mean squared error (MSE), and a custom intensity-weighted MSE (IW-MSE). The ℓ_1 loss minimizes the absolute difference between predicted and ground truth images:

$$\mathcal{L}_{\ell_1}(x, \hat{x}) = \|x - \hat{x}\|_1,$$

while the MSE loss penalizes the squared difference:

$$\mathcal{L}_{\text{MSE}}(x, \hat{x}) = \|x - \hat{x}\|_2^2.$$

To better account for local contrast in scientific imaging, we introduced an intensity-weighted MSE loss defined as:

$$\mathcal{L}_{\text{IW-MSE}}(x, \hat{x}) = \frac{1}{N} \sum_{i=1}^{N} (1 + \alpha \cdot |x_i - \mu_i|) \cdot (x_i - \hat{x}_i)^2,$$

where μ_i is the local average of the ground truth x computed via average pooling, and α controls the sensitivity to local contrast (set to 2.0 in our experiments). This formulation emphasizes accurate reconstruction in high-contrast regions, which are often scientifically meaningful.

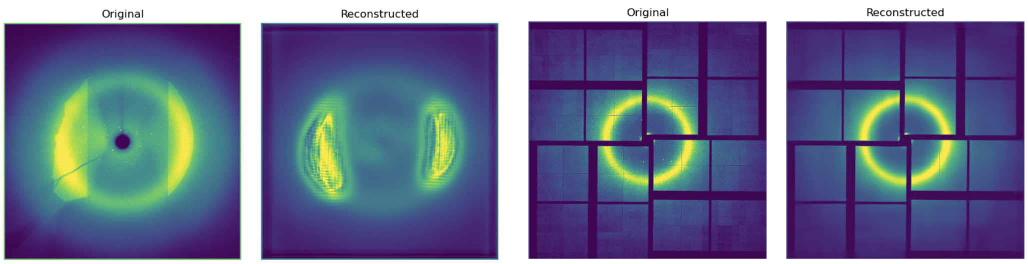


Figure 5: Side-by-side comparison of two original and reconstructed images from two different detectors using VAE with MSE. On the left, we have the detector from experiment 422 and on the right we have detector from experiment 522

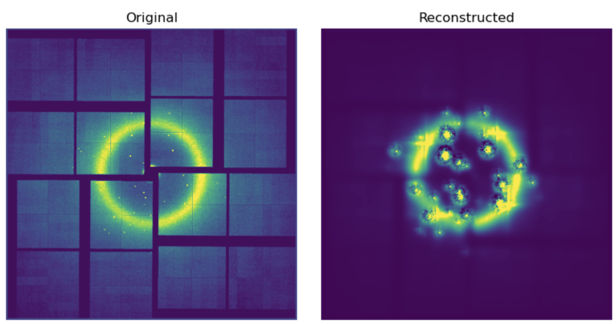


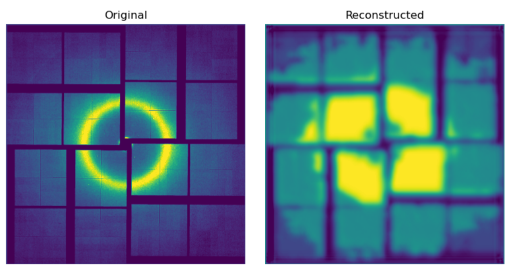
Figure 6: Original and reconstructed image from the detector in experiment 522 using intensity a VAE trained weighted-MSE loss.

For the VAE-KL model, we used a symmetric encoder–decoder architecture composed of four downsampling and upsampling blocks, respectively, using `DownEncoderBlock2D` and `UpDecoderBlock2D`. The architecture operates on single-channel grayscale images and outputs a reconstruction of the same shape. Each stage uses increasing channel widths $(32, 64, 128, 128)$, and a latent bottleneck of 3 channels is used to represent the compressed latent variable z. Additionally, we enabled self-attention in the mid-block to improve global context aggregation. No spatial average pooling was applied unless otherwise noted.

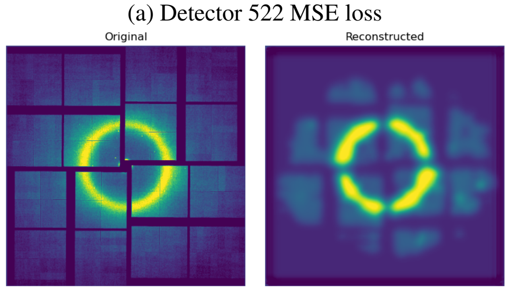
As shown in Figure 5 and Figure 6, we see that, while the overall reconstruction quality of the model trained with intensity-weighted MSE may not visually appear to be the most accurate, it demonstrates superior performance in capturing the fine-grained structure of the Bragg peaks. Compared to standard MSE and L1 loss functions, the intensity-weighted objective places greater emphasis on high-contrast regions, which helps preserve the sharpness and localization of these scientifically meaningful features during reconstruction. Similarly, however, the KL-annealing reconstruction model with MSE does provide a significant benefit in the overall quality of the reconstruction of the image.

4.2 VQ-VAE

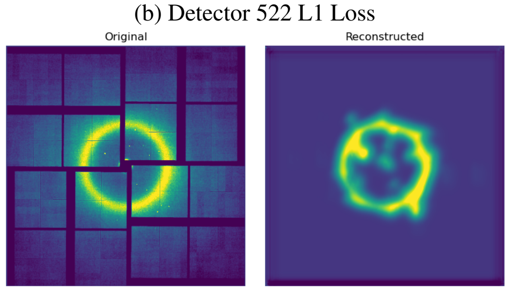
We followed a similar training procedure as in the VAE experiments, using a batch size of 2, weight decay of 0.001, learning rate of 1×10^{-5}, and 10 training epochs. The reconstruction loss used for this model was the intensity-weighted mean squared error (IW-MSE) with an α value of 2.0, placing greater emphasis on reconstructing high-contrast regions. The VQ model utilized a latent bottleneck with a single channel and a spatial encoder–decoder architecture consisting of 4 downsampling and 4 upsampling blocks (`DownEncoderBlock2D` and `UpDecoderBlock2D`, respectively), with output channels set to $(32, 64, 128, 128)$ across the blocks. Each block contains a single layer, and the activation function used is `silu`. To discretize the latent space, we employed a vector quantization layer with a codebook size of 256 and spatial norm regularization. The number of latent embeddings defines the vocabulary over which encoder outputs are quantized, enabling discrete latent representation. The quantizer maps the continuous encoder output to the nearest codebook vector, enforcing a discrete bottleneck during reconstruction. The model was configured with a latent scaling factor of 1, and normalization was performed using 32 groups with spatial normalization.



(a) Detector 522 MSE loss



(b) Detector 522 L1 Loss



(c) Detector 522 with intensity-weighted MSE loss

Figure 7: Comparison of reconstruction results of VQ-VAE model from different detectors and loss functions.

Figure 7 presents a comparison of reconstruction results from the VQ-VAE model using detector 522 under three different loss functions: MSE, L1, and intensity-weighted MSE. The results show that while the model trained with standard MSE loss struggles to reconstruct fine-grained details, particularly in high-contrast areas, it does retain the global structure of the original image. The L1 loss offers slight improvements in the reconstruction of the ring-like features but still lacks precision in capturing subtle intensity variations. In contrast, the model trained with the intensity-weighted MSE loss yields sharper and more localized reconstructions of Bragg peaks, emphasizing high-intensity regions more effectively.

4.3 MAR

To enable efficient training of the MAR model, we employed a two-stage setup where the decoder was initialized from a pre-trained VAE and subsequently frozen during MAR training. This allowed us to train the masked autoregressive model for only 10 epochs while retaining high-quality reconstructions from the latent space. The MAR module was trained to predict continuous latent variables, which were then decoded using the fixed VAE decoder. The diffusion model used to supervise the training followed the Elucidated Diffusion framework and was configured with a minimal number of sampling steps (`num_sample_steps` $= 2$) to reduce computation. We adopted a noise schedule with `sigma_min` $= 0.002$ and `sigma_max` $= 1.2$, and used a standard deviation of `sigma_data` $= 0.5$ to model the underlying data distribution. The sampling behavior was shaped using $\rho = 4$, which controls the interpolation between noise levels, and noise perturbations during training were drawn from a log-normal distribution with mean `P_mean` $= -0.5$ and standard deviation `P_std` $= 0.5$. At inference time, stochastic sampling was enabled with `S_churn` $= 2$, `S_tmin` $= 0.05$, `S_tmax` $= 10$, and `S_noise` $= 1.003$, following the recommendations for stable generation across scales. We set `clamp_during_sampling` to `True` to prevent divergence during noisy sampling steps.

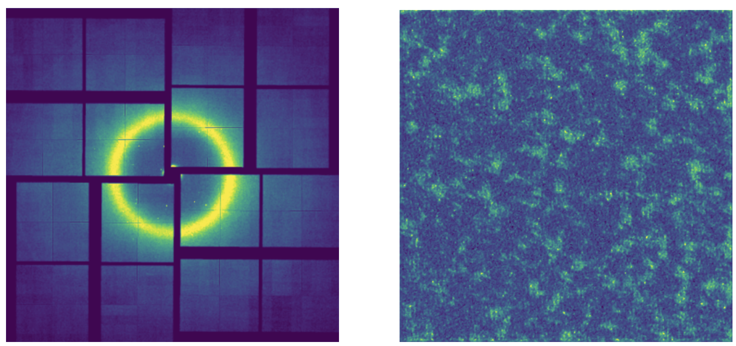
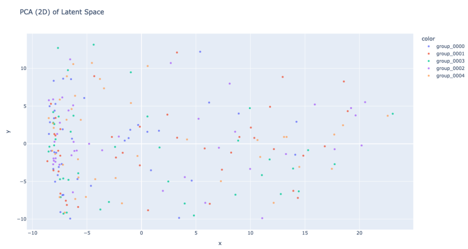


Figure 8: Side-by-side comparison of original and reconstructed image of experiment 522 with diffusion process. It is possible to see that the diffusion process has not been calibrated correctly. This could be related to a very powerful diffusion model that degrades the reconstruction capabilities of VAE.

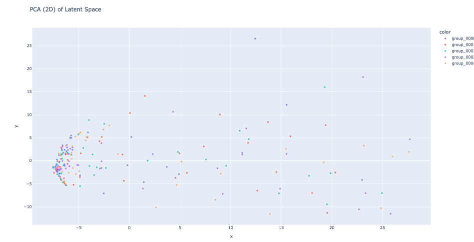
Figure 8 illustrates a reconstruction result using the VAE combined with the diffusion model for experiment 522. While the original image retains a clear ring structure with high-contrast Bragg peaks, the reconstructed output appears overly noisy and unstructured, indicating a failure in preserving the latent semantics necessary for faithful reconstruction. This suggests that the diffusion model may be overly dominant or insufficiently calibrated in its training configuration. When coupled with a frozen VAE decoder, an overpowered diffusion model can easily distort the latent representation in ways that fall outside the decoder's learned distribution, leading to poor reconstructions. To address this, it may be necessary to fine-tune the training parameters of the diffusion model (such as the number of diffusion steps, noise schedule, or loss weighting) to better align with the VAE's latent space constraints and avoid overwhelming its generative capabilities.

4.4 Latent Space Analysis VAE

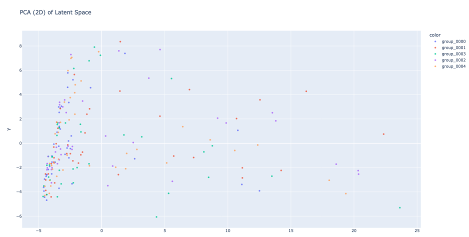
To analyze the latent space, we applied both UMAP and PCA techniques to all the models trained. We used 200 images drawn from five distinct groups as an example set to analyze the latent space. We passed these images through the trained encoder and ,then, extracted the mean of the latent distribution for VAE models and the latent vector for VQ-VAE. Figure 9 presents PCA results for different models. Analysis along the first principal component revealed a strong correlation with the number of Bragg peaks present in the images. This relationship is clearly visualized in Figure 10. Analysis along the second principal component revealed a correlation with the brightness of the images. This relationship is clearly visualized in Figure 11. Interestingly, these characteristics for the first PCA is consistent across different models.



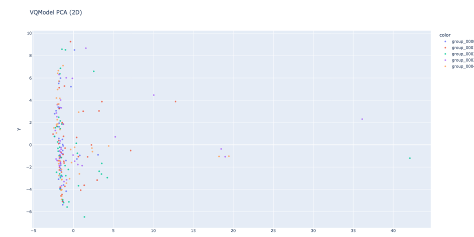
(a) 2D PCA for VAE with Direct Contrast loss



(b) 2D PCA for VAE with Local Contrast loss

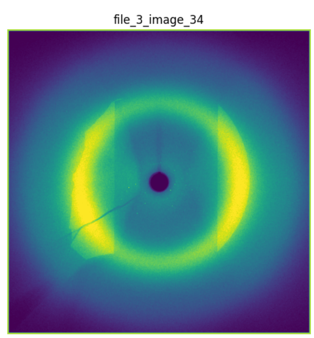


(c) 2D PCA for VAE with MSE loss

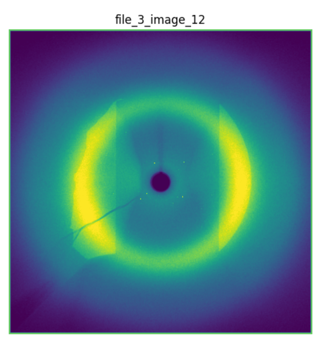


(d) 2D PCA for VQ-VAE

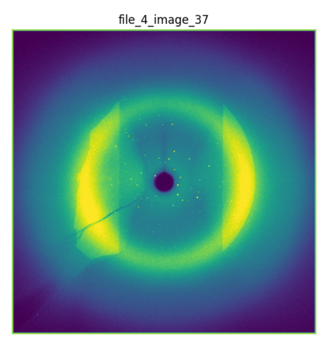
Figure 9: PCA Graph



(a) The image with low first PCA value

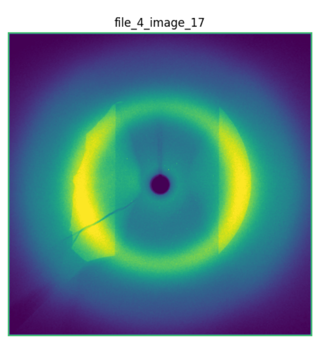


(b) The image with middle first PCA value

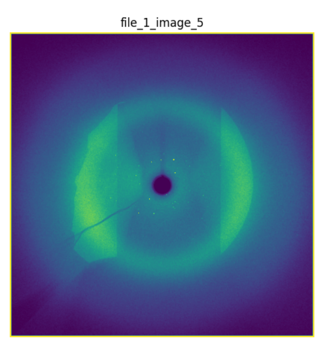


(c) The image with high first PCA value

Figure 10: First PCA Result Analysis



(a) The image with low second PCA value



(b) The image with high second PCA value

Figure 11: Second PCA Result Analysis

5 Conclusion

Our experiments demonstrate that Variational Autoencoders (VAEs), when trained with intensity-weighted MSE loss and KL annealing, are particularly effective at learning the underlying distribution of high-resolution X-ray diffraction images. The use of an intensity-weighted objective emphasizes high-contrast regions, such as Bragg peaks, which are scientifically critical, allowing the model to allocate more representational capacity to informative signal rather than background noise. Meanwhile, KL annealing encourages the model to gradually align the learned latent distribution with the prior, mitigating posterior collapse and enabling more meaningful latent representations. This combination results in a model that not only captures global structure but also retains sufficient sensitivity to rare, high-intensity features, making it well-suited for scientific inference tasks.

In contrast, the VQ-VAE model, while inherently well-matched to the discrete and structured nature of diffraction data, is limited by the fixed capacity of its codebook. A smaller codebook constrains the model's ability to fully express the complexity of the data distribution, particularly in the presence of rare or subtle features. This leads to a trade-off: although VQ-VAEs can offer strong regularization and enable more stable reconstructions, they may underperform in capturing the nuanced variability present in large, diverse datasets unless sufficiently large or adaptive codebooks are used.

The MAR model introduces an autoregressive diffusion-based mechanism for learning latent transitions, aiming to model the conditional structure of latent representations more effectively. However, our observations suggest that the diffusion process, while powerful, can easily overpower the latent space learned by the VAE if not carefully calibrated. In particular, when the MAR module is trained independently and then coupled with a frozen VAE decoder, we observe significant degradation in reconstruction quality—often resulting in outputs that deviate drastically from the true data distribution. This highlights the sensitivity of diffusion-based latent modeling to hyperparameter tuning and initialization. Nonetheless, the MAR architecture shows strong potential for learning flexible and expressive generative models when trained in harmony with the encoder-decoder pipeline. If properly tuned, it could enable structured generation and inpainting within the learned latent space, offering a promising path for further exploration.

Taken together, these findings underscore the critical importance of accurately modeling the underlying data distribution, particularly in the context of sparse and high-resolution X-ray diffraction (XRD) data. Future work should prioritize scaling model capacity and refining training strategies to capture more complex and fine-grained structures without compromising stability or reconstruction quality. This is especially pertinent for architectures like VQ-VAE and MAR, which exhibit strong potential but remain sensitive to design and calibration choices. Moreover, the ability of these models to learn compact latent representations and generate realistic synthetic samples presents valuable opportunities for downstream applications, including efficient storage, accelerated inference, and data augmentation in scientific workflows.

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